Question Paper

Physics Single Correct (Maximum Marks: 80)

Question No. 1

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

An electric dipole has the magnitude of its charge as q and its dipole moment is p. It is placed in a uniform electric field E. If its dipole moment is along the direction of the field, the force on it and its potential energy are respectively

A. 2q.E and minimum

B. q.E and p.E

C. Zero and minimum

D. q.E and maximum

Question No. 2

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

What should be the characteristic of fuse wire?

A. High melting point, high specific resistance.

B. Low melting point, low specific resistance.

C. High melting point, low specific resistance.

D. Low melting point, high specific resistance.

JOIN - @IIT_JEE_NEET_BOOK

Question No. 3

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A charge q is uniformly distributed on a ring of radius r. A sphere of an equal radius is constructed with its centre lying on the periphery of the ring. The flux of electric field through the surface of the sphere will be

A. $\frac{q}{\varepsilon_0}$

B. $\frac{q}{2\varepsilon_0}$

C. $\frac{q}{3\varepsilon_0}$

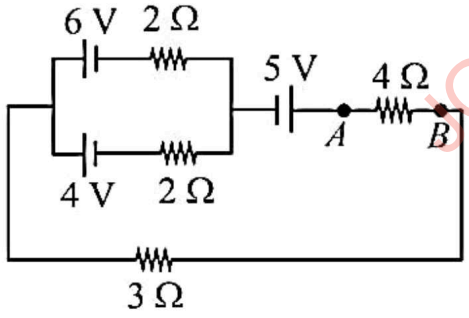
D. $\frac{q}{4\varepsilon_0}$

Question No. 4

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

In the circuit shown (figure), calculate the current through 6 V battery.



A. $(1/4)$ A

B. $(1/8)$ A

C. $(1/2)$ A

D. None of these

JOIN - @IIT_JEE_NEET_BOOK

Question No. 5

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Assertion : The force between the plates of a parallel plate capacitor is proportional to charge on it.

Reason : Electric force is equal to charge per unit area.

A. If both assertion and reason are true and reason is the correct explanation of assertion.

B. If both assertion and reason are true but reason is not the correct explanation of assertion.

C. If assertion is true but reason is false.

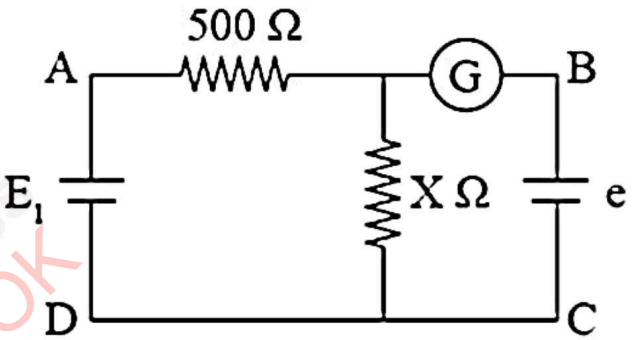
D. If both assertion and reason are false.

Question No. 6

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

In an experiment according to set up, when $E_1 = 12$ V and internal resistance zero, $e = 2$ V. The galvanometer reads zero, then X would be-



A. 200

B. 500

JOIN - @IIT_JEE_NEET_BOOK
u/wrapthatass

C. 100

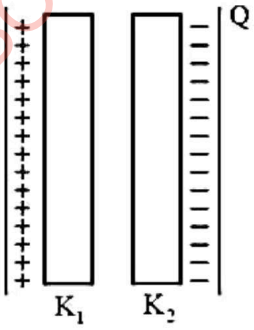
D. 10

Question No. 7

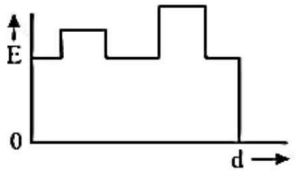
Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

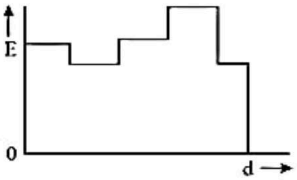
Two thin dielectric slabs of dielectric constants K_1 and $K_2(K_1 < K_2)$ are inserted between plates of a parallel plate capacitor, as shown in the figure. The variation of electric field 'E' between the plates with distance 'd' as measured from plate P is correctly shown by:



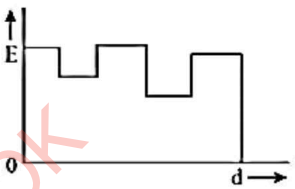
A.



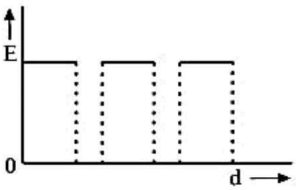
B.



C.



D.



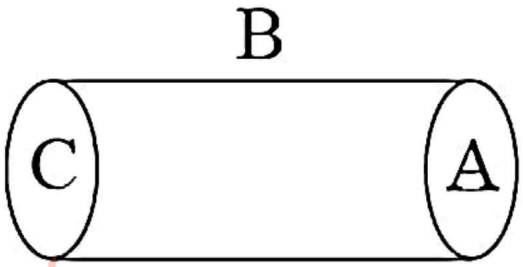
JOIN - @IIT_JEE_NEET_BOOK

Question No. 8

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A hollow cylinder has a charge q placed at its centre. If ϕ is the electric flux (in Vm) associated with the curved surface B, the flux linked with the plane surface A is



A. $\frac{1}{2}\left(\frac{q}{\varepsilon_0}-\phi\right)$

B. $\frac{q}{2\varepsilon_0}$

C. $\frac{\phi}{3}$

D. $\frac{q}{\varepsilon_0}-\phi$

Question No. 9

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A fully charged parallel plate capacitor of $1\mu\mathrm{F}$ capacity is discharging through a resistor. If the energy of the capacitor reduces to half in one second then the value of resistance will be

A. $\frac{2\times10^6}{\ln 2}\ \Omega$

B. $\frac{2\times10^3}{\ln 2}\ \Omega$

C. $\frac{10^6}{\ln 2}\ \Omega$

D. $\frac{10^3}{\ln 2}\ \Omega$

JOIN - @IIT_JEE_NEET_BOOK
u/wrapthatass

Question No. 10

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Column - I gives certain physical terms associated with flow of current through a metallic conductor. Column - II gives some mathematical relations involving electrical quantities. Match Column - I and Column - II with appropriate relations.

Column - I	Column - II
(A) Drift Velocity	$(P)\ \frac{m}{ne^2\rho}$
(B) Electrical Resistivity	$(Q) nev_d$
(C) Relaxation Period	$(R)\ \frac{eE}{m}\tau$
(D) Current Density	$(S)\ \frac{E}{J}$

A. $(A)-(R),(B)-(S),(C)-(P),(D)-(Q)$

B. $(A)-(R),(B)-(S),(C)-(Q),(D)-(P)$

C. $(A)-(R),(B)-(P),(C)-(S),(D)-(Q)$

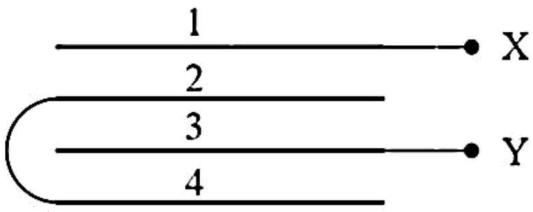
D. $(A)-(R),(B)-(Q),(C)-(S),(D)-(P)$

Question No. 11

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Four metal plates are arranged as shown in the figure. Capacitance between X and Y ($A \rightarrow$ Area of each plate, $d \rightarrow$ distance between the plates) is



A. $\frac{3}{2}\ \frac{\varepsilon_0 A}{d}$

B. $\frac{2\varepsilon_0 A}{d}$

C. $\frac{2}{3}\ \frac{\varepsilon_0 A}{d}$

D. $\frac{3\varepsilon_0 A}{d}$

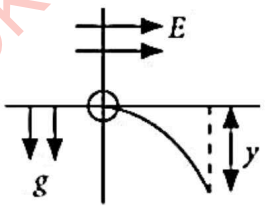
Question No. 12

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A charge particle is moving along electric field as shown in figure. What is horizontal displacement of the charge particle when it descends a distance of y meter?

Given $Q/m = 9.6 \times 10^7 \text{C/kg}$, $E = 5 \times 10^5$ V/m, $y = 84$ cm, $g = 10$ m/s^2



A. 4.03×10^{12} m

B. 4.03×10^{10} m

C. 4.03×10^{-8} m

D. 4.03×10^{8} m

Question No. 13

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A voltmeter of resistance 20000Ω reads 5 volt. To make it read 20 volt, the extra resistance required is

A. 40000Ω in parallel

B. 60000Ω in parallel

C. 60000Ω in series

D. 40000Ω in series

Question No. 14

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A clock face has negative charges $-q, -2q, -3q, \ldots, -12q$ fixed at the position of the corresponding numerals on the dial. The clock hands do not disturb the net field due to point charges. At what time does the hour hand point in the same direction of the electric field at the centre of the dial?

A. $3:30$

B. $9:30$

C. $6:30$

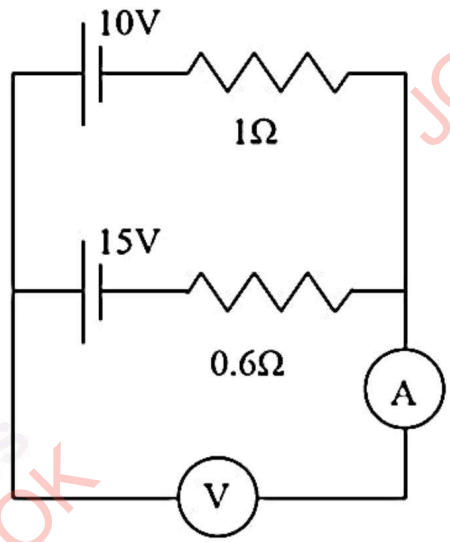
D. $12:30$

Question No. 15

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A 10 V battery with internal resistance 1Ω and a 15 V battery with internal resistance 0.6Ω are connected in parallel to a voltmeter of resistance 1Ω (see figure). The readings in the ideal ammeter is :



A. $\frac{115}{11}\ A$

B. $\frac{105}{11}$ A

C. $\frac{105}{21}$ A

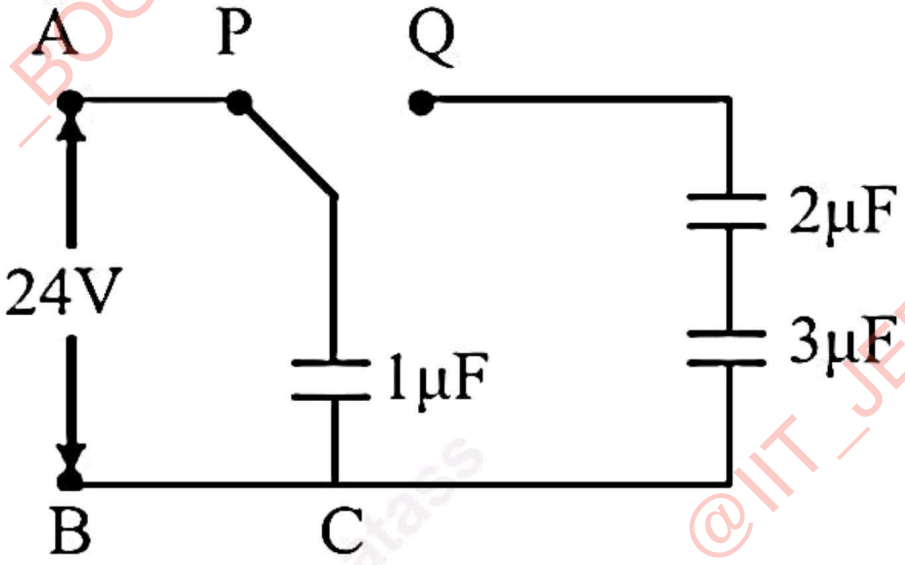
D. $\frac{115}{21}$ A

Question No. 16

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A constant potential difference between A and B is applied. When the switch is shifted from position P to position Q, then



charge on $2\mu F$ capacitor is

A. approximately $13\mu C$

potential difference across $3\mu F$ capacitor is

B. approximately 4.3 V

potential difference across $1\mu F$ capacitor is

C. approximately 11 V

D. All of the above

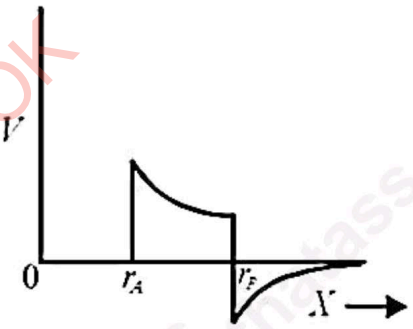
Question No. 17

Only One Option Correct Type

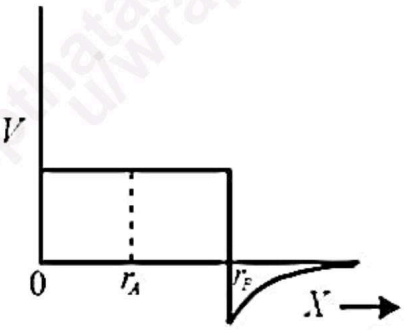
Each question has multiple options out of which ONLY ONE is correct.

Two concentric conducting thin spherical shells A and B having radii r_A and $r_B (r_B > r_A)$ are charged to Q_A and $-Q_B (|Q_B| > |Q_A|)$. The electrical potential along a line, (passing through the centre is)

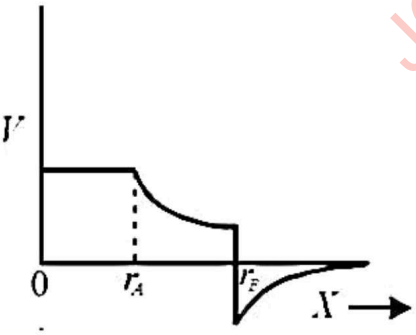
A.



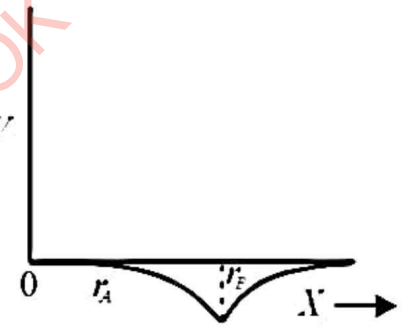
B.



C.



D.



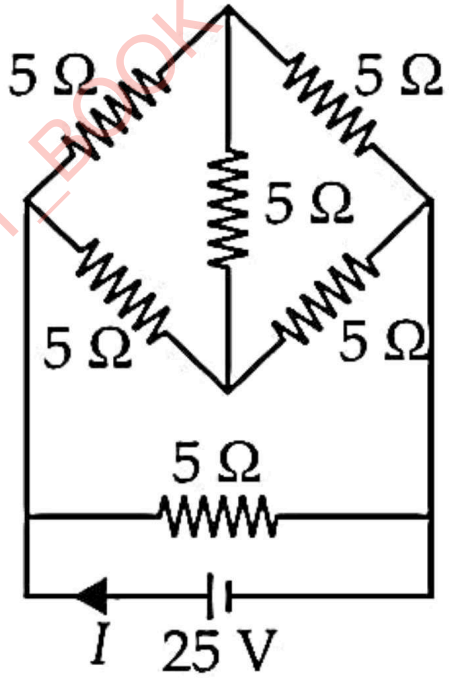
JOIN - @IIT_JEE_NEET_BOOK
u/wraptnatass

Question No. 18

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Calculate I for the given circuit diagram.



A. 10 A

B. 5 A

C. 2.5 A

D. 20 A

JOIN - @IIT_JEE_NEET_BOOK

Question No. 19

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Assertion : A charged particle free to move in an electric field always moves along an electric field line.

Reason: The electric field lines diverge from a negative charge and converge at a positive charge.

A. If both assertion and reason are true and reason is the correct explanation of assertion.

B. If both assertion and reason are true but reason is not the correct explanation of assertion.

C. If assertion is true but reason is false.

D. If both assertion and reason are false.

Question No. 20

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A conducting sphere of radius R carrying charge Q lies inside an uncharged conducting shell of radius $2R$. If they are joined by a metal wire, the amount of heat that will be produced is

A. $\frac{1}{4\pi\varepsilon_0} \cdot \frac{Q^2}{4R}$

B. $\frac{1}{4\pi\varepsilon_0} \cdot \frac{Q^2}{2R}$

C. $\frac{1}{4\pi\varepsilon_0} \cdot \frac{Q^2}{R}$

D. $\frac{2}{4\pi\varepsilon_0} \cdot \frac{Q^2}{3R}$

JOIN - @IIT_JEE_NEET_BOOK

Physics Numerical (Maximum Marks: 20)

Question No. 1

Numerical Type

The answer has to be filled into the input box provided below.

A galvanometer, having a resistance of 50Ω, gives a full scale deflection for a current of 0.05 A. The length in metre of a resistance wire of area of cross-section 2.97×10^{-2} cm^2 that can be used to convert the galvanometer into an ammeter which can read a maximum of 5 A current is :

$\left(\text{Specific resistance of the wire} = 5 \times 10^{-7}\Omega - \text{m}\right)$

Question No. 2

Numerical Type

The answer has to be filled into the input box provided below.

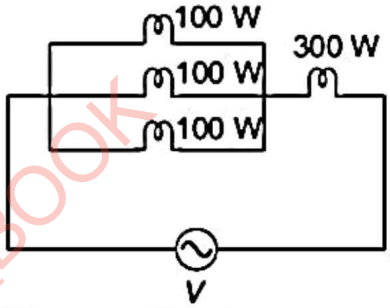
A solid sphere of radius R has a charge Q distributed in its volume with a charge density $\rho = kr^a$, where k and a are constants and r is the distance from its centre. If the electric field at $r = \frac{R}{2}$ is $\frac{1}{8}$ times that at $r = R$, the value of a is

Question No. 3

Numerical Type

The answer has to be filled into the input box provided below.

Four bulbs of different power readings are arranged in a circuit as shown below



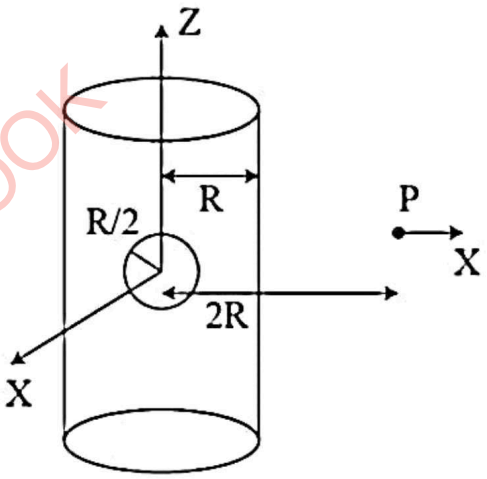
The resultant power of combination if rated voltage is applied across the combination is .....W.

Question No. 4

Numerical Type

The answer has to be filled into the input box provided below.

An infinitely long solid cylinder of radius R has a uniform volume charge density ρ. It has a spherical cavity of radius $R/2$ with its centre on the axis of the cylinder as shown in the figure. The magnitude of the electric field at the point P, which is at a distance 2R from the axis of the cylinder, is given by the expression $\frac{23\rho R}{16K\varepsilon_n}$. The value of K is

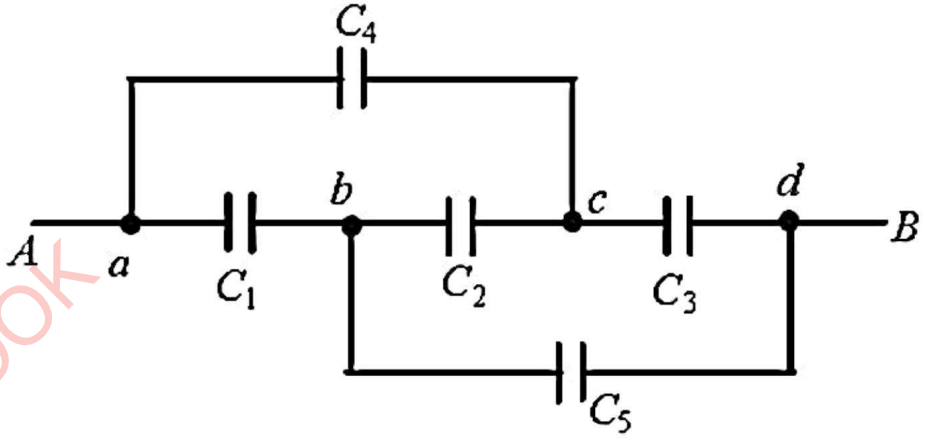


Question No. 5

Numerical Type

The answer has to be filled into the input box provided below.

In the given figure, the capacitance C_1, C_3, C_4, C_5 have a capacitance 4μF each. If



the capacitor C_2 has a capacitance 10μF, then effective capacitance between A and B will be (in Î¼F)

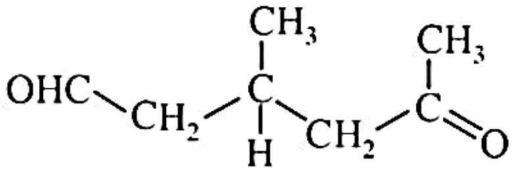
Chemistry Single Correct (Maximum Marks: 80)

Question No. 1

Only One Option Correct Type

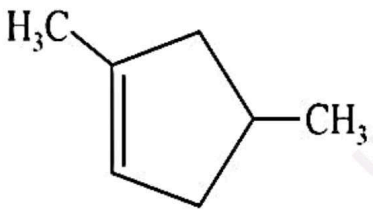
Each question has multiple options out of which ONLY ONE is correct.

A single compound of the structure

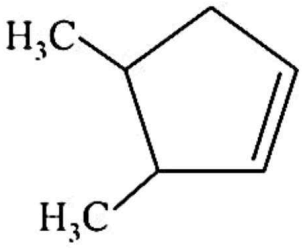


is obtainable from ozonolysis of which of the following cyclic compounds?

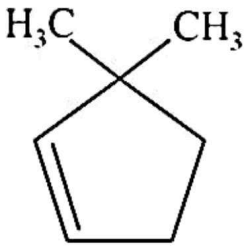
A.



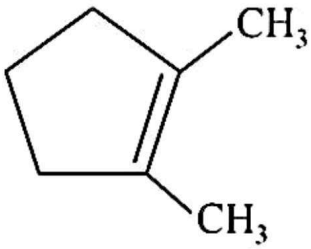
B.



C.



D.



JOIN - @IIT_JEE_NEET_BOOK
u/wrapthatass

Question No. 2

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

An organic compound (A) reacts with sodium metal and forms (B). On heating with conc. H_2SO_4, (A) gives diethyl ether, (A) and (B) are

A. C_2H_5OH and C_2H_5ONa

B. C_3H_7OH and CH_3ONa

C. CH_3OH and CH_3ONa

D. C_4H_9OH and C_4H_9ONa

Question No. 3

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Which of the following is an example of aldol condensation?

A. $2CH_3COCH_3 \xrightarrow{\text{Dil.NaOH}} CH_3COHCH_3CH_2COCH_3$

B. $2HCHO \xrightarrow{\text{Dil.NaOH}} CH_3OH$

C. $C_6H_5CHO + HCHO \xrightarrow{\text{Dil.NaOH}} C_6H_5CH_2OH$

D. None of the above

JOIN - @IIT_JEE_NEET_BOOK

Question No. 4

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Which of the following will undergo oxidation reaction with MnO_2?

A. Hex$-3-$ene$-1-$ol

B. Cyclohex$-3-$ene$-1-$ol

C. Hex$-2-$ene$-1-$ol

D. All of these

Question No. 5

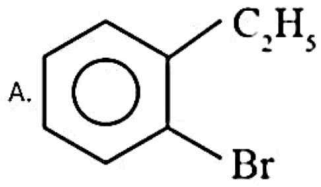
Only One Option Correct Type

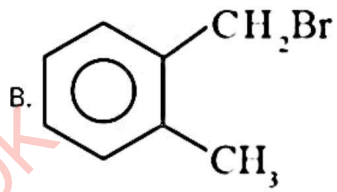
Each question has multiple options out of which ONLY ONE is correct.

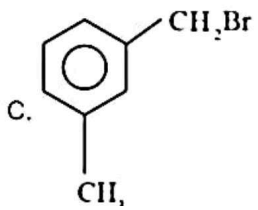
Compound (A), C_8H_9Br gives a pale yellow ppt when warmed with alcoholic $AgNO_3$. Treatment of A with aqueous KOH gave B which reacts as shown below to form phthalaldehyde.

$$B \xrightarrow{MnO_2} \frac{(i)\ CrO_2Cl_2}{(ii)\ H_3O^+} \rightarrow \text{phthalaldehyde}$$

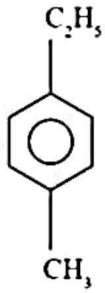
The structure of compound A is:

A. 

B. 

C. 

JOIN - @IIT_JEE_NEET_BOOK

D. 

Question No. 6

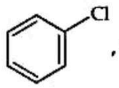
Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Which one of the following reaction sequence is incorrect method to prepare phenol?

A. Aniline, $NaNO_2 + HCl, H_2O,$ heating

B. Cumene, O_2, H_3O^+

C.  , NaOH, STP condition

D.  , oleum, NaOH, H_3O^+

Question No. 7

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Consider the following reaction,

ethanol $\xrightarrow{PBr_3}$ X $\xrightarrow{\text{alc. KOH}}$ Y $\xrightarrow[\text{(ii) } H_2O, \text{ heat}]{\text{(i) } H_2SO_4 \text{ room temperature}}$ Z;

the product Z, is

A. $CH_2 = CH_2$

B. $CH_3CH_2 - O - CH_2 - CH_3$

C. $CH_3 - CH_2 - O - SO_3H$

D. CH_3CH_2OH

JOIN - @IIT_JEE_NEET_BOOK

Question No. 8

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

In the given reaction

$$CH_2 = CH - CHO \xrightarrow{[X]} CH_2 = CH - CH_2OH$$

$[X]$ will be

A. H_2, Ni

B. H_2, Pt

C. $NaBH_4$

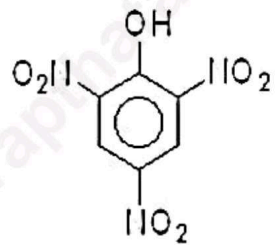
D. H_2/ Wilkinson catalyst

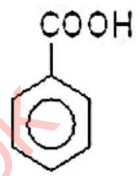
Question No. 9

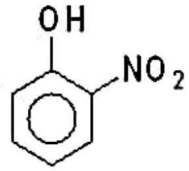
Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

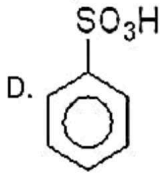
Which of the following is not soluble in $NaHCO_3$ solution?

A. 

B. 

C. 

JOIN - @IIT_JEE_NEET_BOOK
u/wrapthaass

D. 

Question No. 10

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

An aromatic compound 'X' with molecular formula $C_9H_{10}O$ gives the following chemical tests

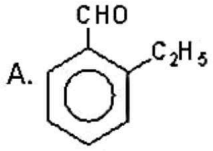
1. Forms 2, 4-DNP derivative,

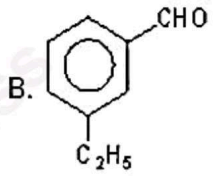
2. Reduces Tollen's reagent,

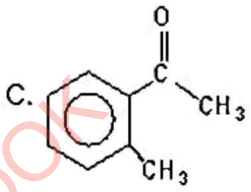
3. Undergoes Cannizaro reaction and,

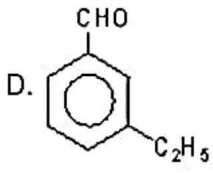
4. On vigorous oxidation 1, 2-benzenedicarboxylic acid is obtained.

X is

A. 

B. 

C. 

D. 

JOIN - @IIT_JEE_NEET_BOOK

Question No. 11

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Phenol on reaction with a mixture of conc. HNO_3 and conc. H_2SO_4 produces a compound. The degree of unsaturation and the nature of compound is

A. 5 , acidic

B. 7 , basic

C. 6 , neutral

D. 7 , acidic

Question No. 12

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Match the compounds given in list I with list II and select the suitable option using the code given belows.

List I		List II	
A.	Benzaldehyde	1.	Phenolphthalein
B.	Phthalic anhydride	2.	Benzoin condensation
C.	Phenyl benzoate	3.	Oil of wintergreen
D.	Methyl salicylate	4.	Fries rearrangement

Code

	A	B	C	D
(a)	4	1	3	2
(b)	4	2	3	1
(c)	2	3	4	1
(d)	2	1	4	3

A. (a)

B. (b)

C. (c)

JOIN - @IIT_JEE_NEET_BOOK
u/wrapthatass

D. (d)

Question No. 13

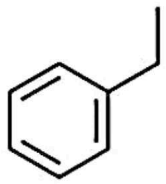
Only One Option Correct Type

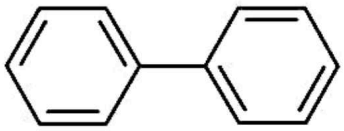
Each question has multiple options out of which ONLY ONE is correct.

Identify the final product [D] obtained in the following sequence of reactions.

$$CH_3CHO \xrightarrow[H_3O^+]{\text{i) } LiAlH_4} [A] \xrightarrow[\Delta]{H_2SO_4} [B] \xrightarrow{HBr} [C] \xrightarrow[\text{Na/Dry ether}]{C_6H_5Br} [D]$$

A. $CH \equiv C^- Na^+$

B. 

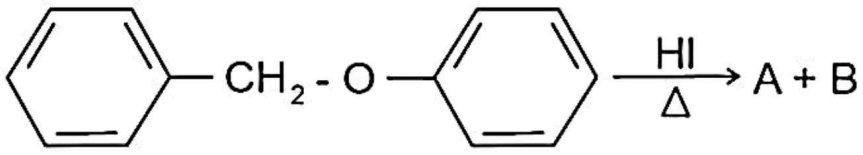
C. 

D. C_4H_{10}

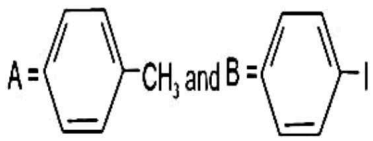
Question No. 14

Only One Option Correct Type

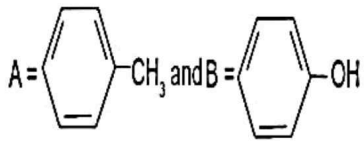
Each question has multiple options out of which ONLY ONE is correct.

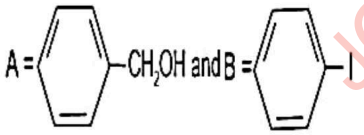


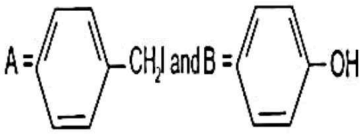
Identify A and B

A. 

JOIN - @IIT_JEE_NEET_BOOK

B. 

C. 

D. 

Question No. 15

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

CH_3CHO and $C_6H_5CH_2CHO$ can be distinguished chemically by

A. Benedict test

B. Iodoform test

C. Tollen's reagent test

D. Fehling solution test

Question No. 16

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

In a set of the given reactions, acetic acid yielded a product 'c'.

$$CH_2COOH + PCl_5 \rightarrow a \xrightarrow[\text{Anh.}AlCl_3]{C_6H_6} b \xrightarrow[\text{ether}]{C_2H_5MgBr} C$$

Product 'c' would be:

A. $CH_3CH(OH)C_2H_5$

B. $CH_3COC_6H_5$

C. $CH_3CH(OH)C_6H_5$

D. $CH_3 - \underset{\displaystyle C_2H_5}{\underset{|}{C}}(OH)C_6H_5$

Question No. 17

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Given below are two statements:

Statement I: The acidic strength of mono substituted nitrophenol is higher than phenol because of electron withdrawing nitro group.

Statement II: o-nitrophenol, m-nitrophenol and p-nitrophenol will have same acidic strength as they have one nitro group attached to the phenolic ring.

In the light of the above statements, choose the most appropriate answer from the options given below :

A. Both Statement I and Statement II are incorrect.

B. Statement I is correct but Statement II is incorrect.

C. Statement I is incorrect but Statement II is correct.

D. Both Statement I and Statement II are correct.

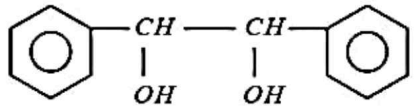
JOIN - @IIT_JEE_NEET_BOOK

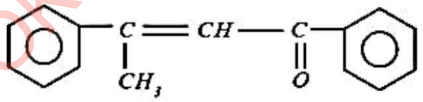
Question No. 18

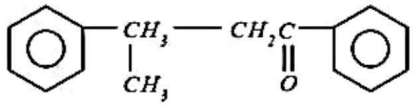
Only One Option Correct Type

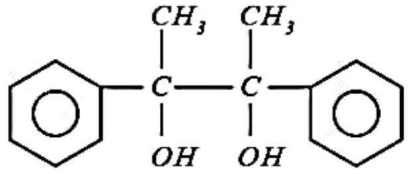
Each question has multiple options out of which ONLY ONE is correct.

Acetophenone when reacted with a base, C_2H_5ONa, yields a stable compound which has the structure?

A. 

B. 

C. 

D. 

Question No. 19

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Which of the following reagents would distinguish cis-cyclopenta-1, 2-diol from the trans-isomer?

A. Acetone

B. Ozone

C. MnO_2

D. Aluminium isopropoxide

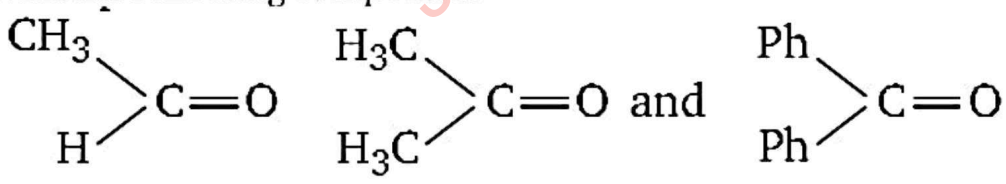
JOIN - @IIT_JEE_NEET_BOOK

Question No. 20

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The order of reactivity of phenyl magnesium bromide (PhMgBr) with the following compounds



(I) (II) (III)

A. III > II > I

B. II > I > III

C. I > III > II

D. I > II > III

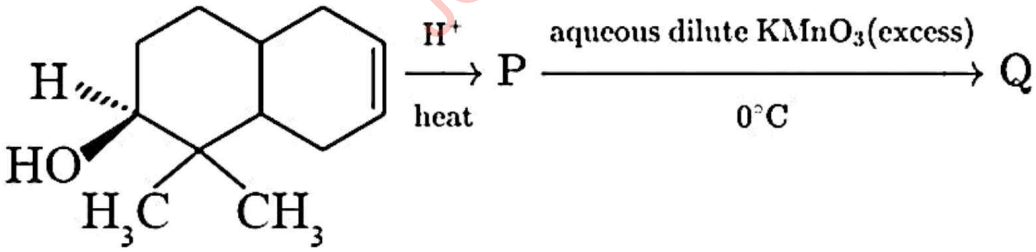
Chemistry Numerical (Maximum Marks: 20)

Question No. 1

Numerical Type

The answer has to be filled into the input box provided below.

The number of hydroxyl group(s) in Q is



Question No. 2

Numerical Type

The answer has to be filled into the input box provided below.

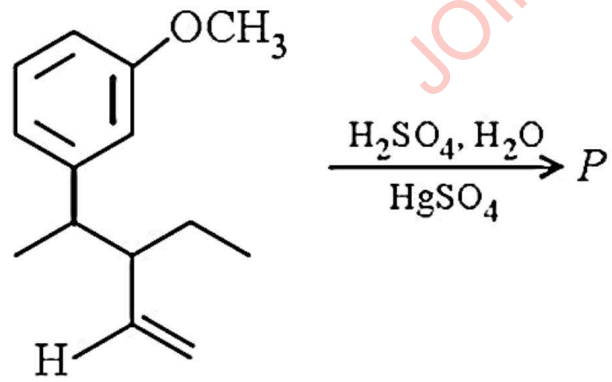
Number of isomeric alcohols of molecular formula $C_6H_{14}O$ which give positive iodoform test is:

JOIN - @IIT_JEE_NEET_BOOK

Question No. 3

Numerical Type

The answer has to be filled into the input box provided below.

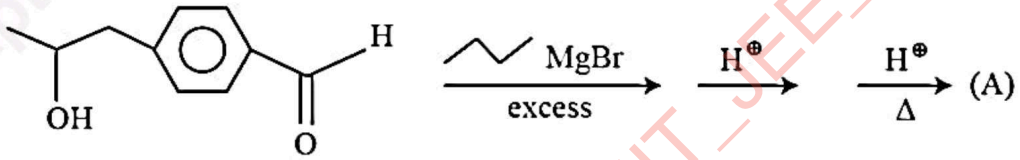


The product P is having how many chiral centres?

Question No. 4

Numerical Type

The answer has to be filled into the input box provided below.



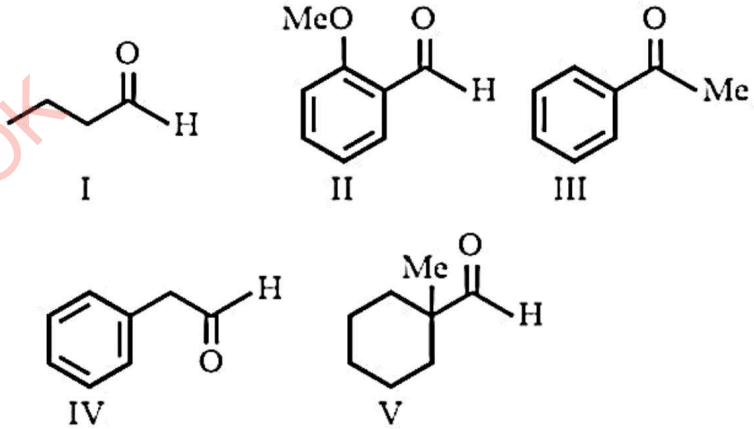
How many geometrical isomer of (A) is possible?

Question No. 5

Numerical Type

The answer has to be filled into the input box provided below.

Among the following,



JOIN - @IIT_JEE_NEET_BOOK
u/wrapthatass

the compounds which can be reduced with formaldehyde and conc. aq. KOH, are:

Mathematics Single Correct (Maximum Marks: 80)

Question No. 1

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Let $f : R \to R$ be defined as $f(x) = \begin{cases} -\frac{4}{3}x^3 + 2x^2 + 3x, & x > 0 \\ 3xe^x \quad , & x \leq 0 \end{cases}$. Then f is increasing function in the interval

A. $\left(-\frac{1}{2}, 2\right)$

B. $(0, 2)$

C. $\left(-1, \frac{3}{2}\right)$

D. $(-3, -1)$

Question No. 2

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

If $f(x) = \begin{cases} \sin x, & x < 0 \\ \cos x - |x-1|, & x \geq 0 \end{cases}$, then $g(x) = f(|x|)$ is non-differentiable for

A. all values of x

B. 3 values of x

C. exactly one value of x

D. none of these

JOIN - @IIT_JEE_NEET_BOOK
u/wrapthatass

Question No. 3

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Let $f(x) = (x-1)^2(x-2)^3e^x$. Then

A. f(x) has local maximum at $x = 1$

B. f(x) has point of inflexion at $x = 1$

C. f(x) has local minimum at $x = 2$

D. all of these

Question No. 4

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Let $a > 0$ be a real number. Then the limit $\lim_{x\to 2} \frac{a^x + a^{3-x} - (a^2+a)}{a^{3-x} - a^{x/2}}$ is

A. $2 \log a$

B. $-\frac{4}{3}a$

C. $\frac{a^2+a}{2}$

D. $\frac{2}{3}(1-a)$

JOIN - @IIT_JEE_NEET_BOOK

Question No. 5

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Number of point(s) in [1,2] where $f(x) = (-2)^{[x^3]}$ is non differentiable (where [.] denotes greatest integer function)

A. 6

B. 5

C. 9

D. 7

Question No. 6

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Let $f : R \to R$ be a differentiable function given by $f(x) = x^3 - 3x + 2020$ If $g(x)$ is a continuous function defined by

$$g\left(x\right) = \begin{cases} \text{Minimum}\{f(t), 0 \le t \le x\}, & 0 \le x \le 1 \\ \text{Maximum}\{f(t), 1 < t \le x\}, & 1 < x \le 2 \end{cases}$$

and m and M be the least and the greatest value of $g(x)$ on $[0, 2]$ then which one of the following is correct?

A. $M - m = 2$

B. $m = 2020$

C. $M = 2022$

D. $m = 2019$

JOIN - @IIT_JEE_NEET_BOOK

Question No. 7

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Let $f(x) = x|x|,\ g(x) = \sin x$ and $h(x) = (gof)(x)$. Then

A. $h(x)$ is differentiable at $x = 0$, but $h'(x)$ is not continuous at $x = 0$

B. $h(x)$ is not differentiable at $x = 0$

C. $h'(x)$ is differentiable at $x = 0$

D. $h'(x)$ is continuous at $x = 0$ but is not differentiable at $x = 0$

Question No. 8

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The value of $\lim_{x\to\infty}\left[\frac{1^{\frac{1}{x}}+2^{\frac{1}{x}}+3^{\frac{1}{x}}+\ldots+10^{\frac{1}{x}}}{10}\right]^{10x}$ is

A. $10!$

B. 10

C. $9!$

D. 0

JOIN - @IIT_JEE_NEET_BOOK

Question No. 9

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Let $f(x)$ be a non-negative differentiable function on $[0, \infty)$ such that $f(0) = 0$ and $f(x) \leq 2f(x)$ for all $x > 0$. Then, on $[0, \infty)$

A. $f(x)$ is always a constant function

B. $f(x)$ is strictly increasing

C. $f(x)$ is strictly decreasing

D. $f'(x)$ changes sign

Question No. 10

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The value of $\lim_{n\to\infty}\left(\frac{[1^2x+1^2]+[2^2x+2^2]+...+[n^2x+n^2]}{n^3}\right)$ is: (where $[\cdot]$ denotes the greatest integer function.)

A. $\frac{x}{3}$

B. $x+\frac{1}{3}$

C. $\frac{x}{3}+\frac{1}{3}$

D. None of these

JOIN @IIT_JEE_NEET_BOOK

Question No. 11

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A 2m ladder leans against a vertical wall. If the top of the ladder begins to slide down the wall at the rate 25cm/sec , then the rate (in cm/sec.) at which the bottom of the ladder slides away from the wall on the horizontal ground when the top of the ladder is 1 m above the ground is:

A. 25

B. $25\sqrt{3}$

C. $\frac{25}{3}$

D. $\frac{25}{\sqrt{3}}$

Question No. 12

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

How many real solutions does the equation $x^7 + 14x^5 + 16x^3 + 30x - 560 = 0$ have?

A. 5

B. 7

C. 1

D. 3

JOIN - @IIT_JEE_NEET_BOOK

Question No. 13

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Number of points of discontinuity of $f(x) = \left[\sin^{-1}x\right] - [x]$ in its domain is equal to (where $[.]$ denotes the greatest integer function)

A. 0

B. 1

C. 2

D. 3

Question No. 14

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Minimum integral value of k for which the equation $e^x = kx^2$ has exactly three real distinct solutions is:

A. 3

B. 0

C. 1

D. 2

JOIN @IIT_JEE_NEET_BOOK

Question No. 15

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The limit of $\left[\frac{1}{x^2}+\frac{(2013)^x}{e^x-1}-\frac{1}{e^x-1}\right]$ as $x \to 0$

A. approaches $+\infty$

B. approaches $-\infty$

C. is equal to $\log_e$, (2013)

D. does not exist

Question No. 16

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Let V_1 be the volume of a given right circular cone with O as the centre of the base and A as its apex. Let V_2 be the maximum volume of the right circular cone inscribed in the given cone whose apex is O and whose base is parallel to the base of the given cone. Then the ratio V_2/V_1 is-

A. $\frac{3}{25}$

B. $\frac{4}{9}$

C. $\frac{4}{27}$

D. $\frac{8}{27}$

JOIN @IIT_JEE_NEET_BOOK
u/wrapthatass

Question No. 17

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

For a real number x let $[x]$ denote the largest number less than or equal to x. for $x \in R$ let $f(x) = [x] \sin \pi x$. Then:

A. f is differentiable on R

B. f is symmetric about the line $x = 0$

C. $\int_{-3}^{3} f(x)dx = 0$

D. For each real α, the equation $f(x) - \alpha = 0$ has infinitely many roots.

Question No. 18

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The set of all values of a for which the function $f(x) = \left(a^2 - 3a + 2\right)\left(\cos^2 x/4 - \sin^2 x/4\right) + (a-1)x + \sin 1$ does not possess critical points is

A. $[1, \infty)$

B. (0,1)$\cup(1,4)$

C. (-2,4)

D. (1,3)$\cup(3,5)$

JOIN - @IIT_JEE_NEET_BOOK

Question No. 19

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

If $\tan^2\alpha + \cot^2\alpha = \lim_{a\to\infty}\left(\sqrt{a^2+8a+3} - \sqrt{a^2+4a+2}\right)$ where $\alpha \in R$, then the value of $\sum_{n=1}^{\infty}\left(\sin^{2n}\alpha + \cos^{2n}\alpha\right)$ is

A. 2

B. 2.5

C. 3

D. 1.5

Question No. 20

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Consider the function

$$f(x) = \begin{cases} \frac{x+5}{x-2} & \text{if } x \neq 2 \\ 1 & \text{if } x = 2 \end{cases}$$

Then $\mathrm{f(f(x))}$ is discontinuous

A. at all real numbers

B. at exactly two values of x

C. at exactly one value of x

D. at exactly three values of x

JOIN - @IIT_JEE_NEET_BOOK

Mathematics Numerical (Maximum Marks: 20)

Question No. 1

Numerical Type

The answer has to be filled into the input box provided below.

$\lim_{h\to\infty} h^{-h^2}\left[(h+1)\left(h+\frac{1}{2}\right)\left(h+\frac{1}{2^2}\right)\ldots\ldots\ldots\ldots\ldots\left(h+\frac{1}{2^{h-}}\right.\right.$
then $S=$.....

Question No. 2

Numerical Type

The answer has to be filled into the input box provided below.

The number of real values of x that satisfies the equation $x^4+4x^3+12x^2+7x-3=0$ is

Question No. 3

Numerical Type

The answer has to be filled into the input box provided below.

If $f(x)=\max\left\{\frac{x}{n},\ |\sin\pi x|\right\},\ n\in N$ has maximum points of non-differentiability for $x\in(0,\ 4)$, then minimum value of n is

JOIN - @IIT_JEE_NEET_BOOK

ıatics Numerical (Maximum Marks: 20)

No. 1

Numerical Type

The answer has to be filled into the input box provided below.

$$\lim_{h\to 0} h^2\left[(h+1)\left(h+\frac{1}{2}\right)\left(h+\frac{1}{2^2}\right)\ldots\ldots\ldots\ldots\ldots\ldots\left(h+\frac{1}{2^{h-1}}\right)\right]^h = e^s$$
$=$.....

No. 2

Numerical Type

The answer has to be filled into the input box provided below.

mber of real values of x that satisfies the equation $x^3+12x^2+7x-3=0$ is

No. 3

Numerical Type

The answer has to be filled into the input box provided below.

$=\max\left\{\frac{x}{n},\ |\sin\pi x|\right\},\ n\in N$ has maximum points of fferentiability for $x\in(0,\ 4)$, then minimum value of n is

Question No. 4

Numerical Type

The answer has to be filled into the input box provided below.

Let $f(x)$ be a function defined by $f(x)=\int_3^x t\left(t^2-7t+12\right)dt, 3 \leq x \leq 6$. If the range of $f(x)$ is $[a, b]$, then the value of $\left(3a+\frac{b}{9}\right)$ is

Question No. 5

Numerical Type

The answer has to be filled into the input box provided below.

The number of points at which the function $f(x)=(x-|x|)^2(1-x+|x|)^2$ is not differentiable in the interval $(-3,4)$ is

JOIN - @IIT_JEE_NEET_BOOK